

Trainings ▾

General Information

Installations can vary from original equipment manufacturer (OEM) to OEM. Use the following steps as a guideline. Reference the OEM manual when necessary.

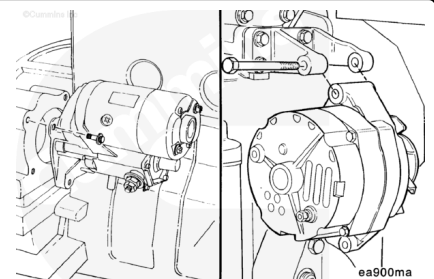
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Install

Install all accessories and brackets that were removed from the engine during engine removal.

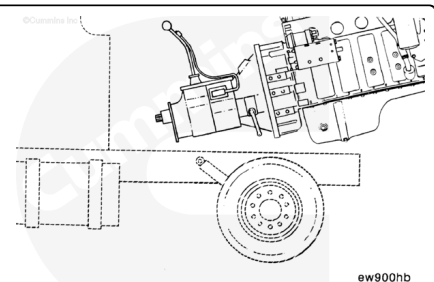
For engine supplied components, reference the corresponding procedure in this manual.

For vehicle installed components, reference the corresponding equipment manufacturer service information.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission safely as an assembly without causing personal injury. For engine weight specifications refer to **Procedure 018-015 (</qs3/pubsys2/xml/en/procedures/40/40-018-015-tr.html>), **General Engine Specifications** in Section V. Reference the equipment manufacturer's specifications for the transmission weight.**



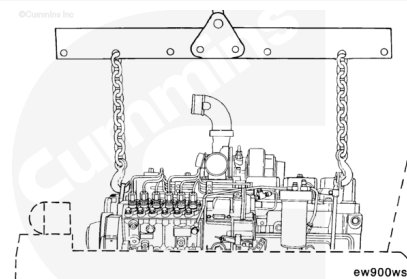
Note : On applications where the rear-engine mounts are attached to the transmission, it is often necessary to install the engine and transmission as an assembly.

Refer to the equipment manufacturer service information.

Use a properly rated hoist and engine-lifting bracket, Part Number 3822512, attached to the engine-mounted lifting brackets to install the engine.

Refer to Procedure 018-015

(</qs3/pubsys2/xml/en/procedures/40/40-018-015-tr.html>), General Engine Specifications in Section for engine weight specifications.

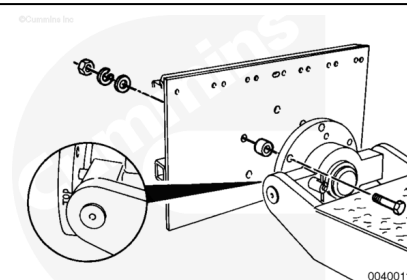


If the engine was repaired, lift the engine and engine stand adapter plate from the engine rebuild stand.

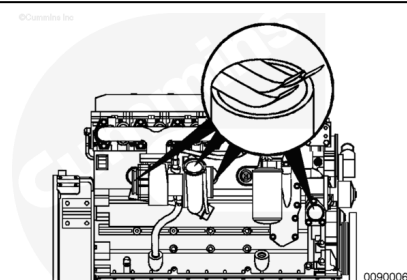
Remove the engine stand adapter plate.

Note : The turbocharger may need to be installed after the engine stand adapter plate is removed. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html>).

If the engine was replaced, lift the engine from the shipping pallet.



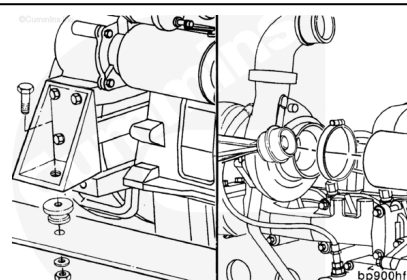
Remove the covers from all openings that were covered during removal.



Align the engine in the chassis, and tighten the engine-mounting capscrews. Refer to the equipment manufacturer's torque specifications.

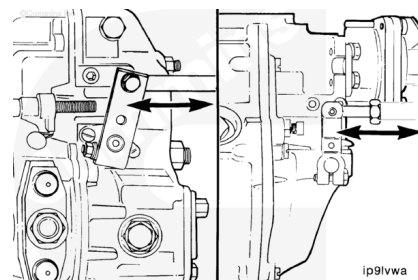
Connect all engine- and chassis-mounted accessories that were removed.

Note : Make sure that all lines, hoses, and tubes are properly routed and fastened to prevent damage. Make sure that the air intake and exhaust pipe connections are tight and free of leaks.

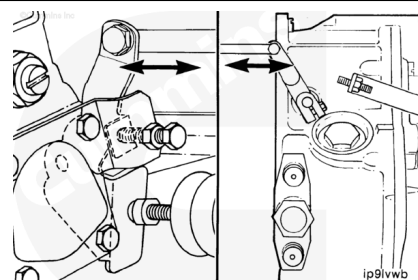


Throttle Control Lever Connection

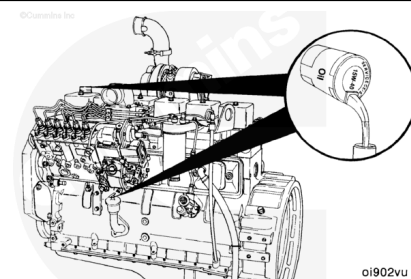
Install and adjust the throttle linkage to the injection pump control lever. Refer to the equipment manufacturer service information.



Adjust the length of the cable/rod to the mechanical shutdown lever so there is stop-to-stop movement. Refer to the equipment manufacturer service information.



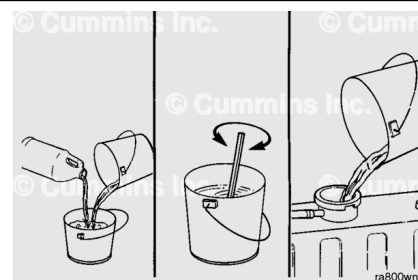
Fill the engine with clean 15W-40 engine oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html).



Note : The total coolant capacity of the engines varies. Refer to the original equipment manufacturer's instructions to determine the capacity of the whole cooling system.

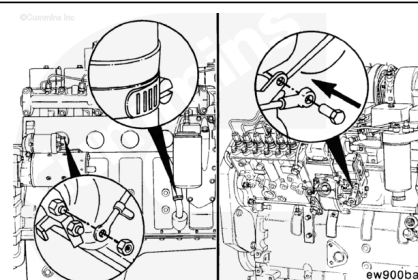
Fill the cooling system with new coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).

Refer to Procedure 018-018 (/qs3/pubsys2/xml/en/procedures/40/40-018-018-tr.html) for cooling system specifications.



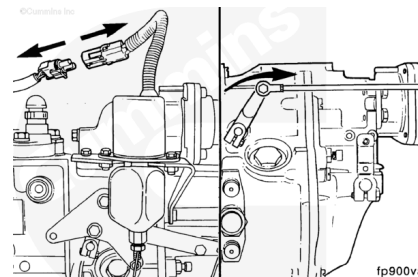
⚠ CAUTION ⚠

Installation of the radiator cap at this point is critical to proper purging of air trapped in the cooling system. Improper purging of air from the cooling system can result in engine damage from overheating.



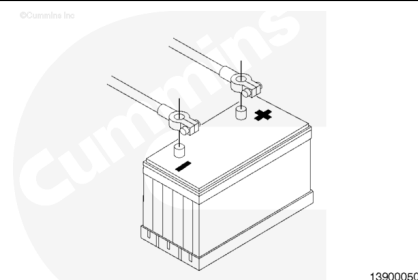
Perform a final inspection to make sure that all hoses, wires, linkages, and components have been properly installed and tightened.

Make sure that the fuel is shut off by removing the wire from the fuel solenoid or by verifying that the mechanical fuel shutoff is in the OFF position. This is necessary to prevent the engine from starting during the lubricating oil rifle pressure charging operation.



⚠ WARNING ⚠

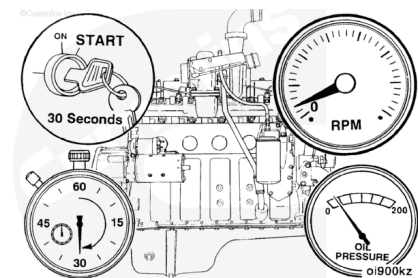
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries.

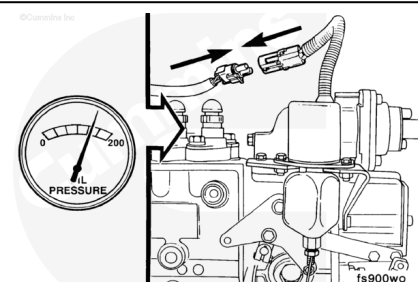
⚠ CAUTION ⚠

Do not engage the starting motor for more than 30 seconds. Wait 2 minutes between starter engagements to cool the starting motor.



Crank the engine until the lubricating oil pressure gauge indicates a positive pressure.

After pressure is observed, connect the wire to the fuel solenoid.

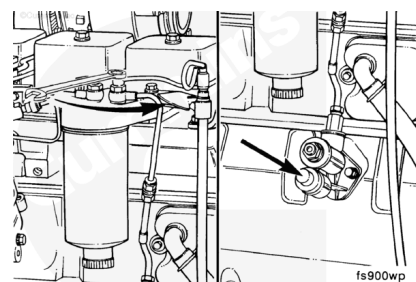


Prime the low-pressure fuel system by operating the plunger on the fuel transfer pump.

If equipped, open the bleed screw and operate the plunger until fuel flowing from the bleed screw is free of air.

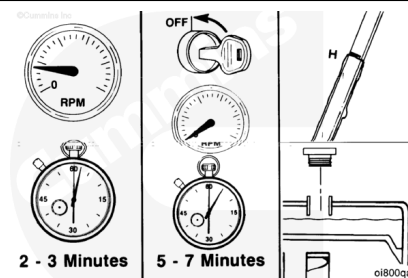
Tighten the bleed screw.

Torque Value: 9 n•m [80 in-lb]



⚠ WARNING ⚠

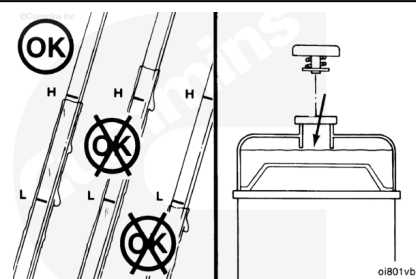
Do not remove the radiator cap from a hot engine. Wait until the engine coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine at low idle for 2 to 3 minutes.

Shut off the engine, and wait 5 to 7 minutes for the lubricating oil to drain to the lubricating oil pan, and check the lubricating oil and coolant levels again.

Fill the engine to the correct lubricating oil and coolant levels, if necessary.



Operate the engine at 1000 to 1200 rpm for 8 to 10 minutes. Check for proper operation, unusual noises, and coolant, fuel, or lubricating oil leaks.

Repair all leaks and component problems. See the appropriate procedures.

See Section 14 for the Engine Run-in and Test Procedures.

